

AI Helps Those Who Have Less? Social Support Gaps and Perceptions of AI Emotional Support

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Abstract

AI emotional support chatbots (e.g., LLMs) promise accessible mental health support, yet how users' social contexts influence their perceptions remains underexplored. We conducted a 10-day diary study (N=24) with an LLM-based chatbot, followed by in-depth interviews (n=8). Using grounded theory, we identified a "support gap" framing: participants with limited social support (e.g., fear of burdening others, unsatisfying relationships) evaluated the AI more positively, viewing it as a judgment-free resource. In contrast, those with strong support networks were more critical, using high-quality human empathy as their reference standard. Our findings suggest that AI evaluation is relative to users' pre-existing social experiences rather than system quality alone. We invite the CHI community to consider how social context should inform the design and ethical evaluation of AI emotional support systems.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; *Natural language interfaces*.

Keywords

AI emotional support, social support, chatbots, mental health, diary study

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1 Introduction

The rise of large language models (LLMs) has sparked growing interest in AI-powered emotional support tools. From mental health chatbots to AI companions, these systems promise accessible, always-available, and non-judgmental support for users navigating stress, anxiety, and everyday emotional challenges. Early research demonstrates promising outcomes: Fitzpatrick et al. [2] found that Woebot, a chatbot grounded in cognitive behavioral therapy, reduced users' anxiety symptoms within two weeks. Inkster et al. [6] showed that Wysa, an empathy-driven conversational agent, helped frequent users reduce depressive symptoms. Yet, as these tools become increasingly integrated into everyday life, a critical question remains underexplored: for whom does AI emotional support work well, and for whom does it fall short?

Current research on AI emotional support has largely focused on system performance metrics, such as empathy perception, response quality, or therapeutic outcomes, often evaluated at an aggregate level. Glikson and Woolley [3] reviewed empirical research on human trust in artificial intelligence, providing a foundation for understanding how users interact with these systems. Hatch et al. [4] found that ChatGPT can pass a Turing test for emotional understanding in ways that resemble human therapists. This approach implicitly assumes that the quality of AI emotional support can be assessed independently of users' backgrounds. However, we propose that users do not evaluate AI support in a vacuum; instead, they judge it against a reference standard shaped by their existing social support experiences. A user whose friends are readily available and empathetic may compare the AI to that high standard and find it lacking, whereas a user who fears burdening others or

has had negative counseling experiences may compare the AI to having no support at all and find it valuable. We call this lens a “support gap” framing: the perceived value of AI emotional support may depend not on system quality alone, but on whether users experience gaps in accessing quality human support.

Recent longitudinal studies offer indirect support for this framing. Zhang et al. [10] found that intense AI companion use was associated with lower well-being, particularly among users with smaller social networks. Fang et al. [1] observed that heavy AI usage heightened emotional dependence for some users. However, these studies primarily examined network size or usage intensity as objective or behavioral variables. They do not address a more subjective question: how do users’ perceived barriers to accessing support, such as fear of burdening others or past negative experiences, shape the evaluative lens through which they judge AI? Rather than measuring support quantity, our study explores how users’ sense of support accessibility and quality may function as a reference standard against which AI is compared.

We thus report an emergent finding from a 10-day diary study in which 24 participants interacted daily with an LLM-based emotional support chatbot, followed by in-depth interviews with 8 participants. Using grounded theory analysis, we observed a consistent pattern: users’ evaluations of AI emotional support varied based on perceived gaps in their pre-existing social support. Participants who described barriers to accessing human support, such as fear of burdening friends, dissatisfaction with existing relationships, or difficulty discussing certain topics, tended to describe the AI as a judgment-free space that addressed unmet needs. In contrast, participants with strong, accessible support networks evaluated the AI against a higher standard and were more likely to find it lacking. As one participant with professional counseling training and a supportive partner noted: “No matter what tool it is, it cannot compare to a real person. My support system is strong enough”.

We hope this study offers an empirical starting point that raises critical questions for the CHI community: How should users’ social contexts inform the design and evaluation of AI emotional support? Should these systems be tailored to users with different support backgrounds? What are the ethical implications of positioning AI as a supplement for those underserved by human support? Through this study, we propose a “support gap” framing, suggesting that users’ evaluations of AI emotional support are shaped by their pre-existing social support experiences rather than system quality alone. This framing invites the CHI community to reconsider how social context should inform the design, evaluation, and ethical positioning of AI emotional support systems, while opening space for further empirical and conceptual exploration.

2 Related Work

2.1 AI Emotional Support: Efficacy and User Perception

A growing body of research has examined the efficacy of AI-powered emotional support tools. Chatbots grounded in therapeutic frameworks, such as Woebot (cognitive behavioral therapy) and Wysa (empathy-driven design), have demonstrated positive outcomes, including reduced anxiety and depressive symptoms [2, 6]. Beyond symptom reduction, research has explored how users perceive AI

empathy. Hatch et al. [4] found that ChatGPT can simulate emotional understanding in ways that resemble human therapists, while Pentina et al. [8] showed that anthropomorphic cues in chatbots can foster emotional attachment and trust. However, most studies evaluate AI performance at an aggregate level, measuring average user satisfaction or therapeutic outcomes without examining how individual differences in users’ backgrounds shape their perceptions. This leaves open a critical question: do all users perceive and evaluate AI emotional support in the same way, or do pre-existing factors influence how users judge these systems?

2.2 Social Support Contexts and Technology Use

House et al. [5] identified multiple mechanisms through which support is delivered, including emotional, informational, and appraisal support. Research in human-computer interaction has begun to consider how users’ existing social contexts influence their engagement with supportive technologies. Zhang et al. [10] found that users with smaller social networks experienced lower well-being from intensive AI companion use. Similarly, Fang et al. [1] observed that heavy AI usage heightened emotional dependence, particularly for users who engaged in high levels of self-disclosure. These findings indicate that the relationship between AI use and user outcomes is not uniform but varies based on users’ social circumstances. However, prior work has primarily examined social support in terms of network size or behavioral indicators, and less attention has been paid to users’ subjective perceptions of their support quality and accessibility, as well as how these perceptions might shape the evaluative lens through which users judge AI emotional support. Our study addresses this gap.

3 Method

3.1 Study Design

We conducted a 10-day diary study to explore users’ experiences with an LLM-based emotional support chatbot. The study employed a qualitative-dominant mixed-methods design, combining daily chatbot interactions, periodic questionnaires, and post-study in-depth interviews. This study was reviewed and approved by the Institutional Review Board (IRB) of the authors’ institution (name anonymized for review).

3.2 Participants

We recruited 27 participants through convenience sampling. All participants completed a pre-screening using a stress scale [7] to ensure they were within an acceptable psychological state (stress index 4–5 on a 10-point scale). To align with the non-clinical nature of the chatbot and to minimize risk, participants with diagnosed mental or psychological conditions were excluded from this study.

Of the 27 enrolled participants, 24 completed all daily interaction tasks and questionnaires, forming our final valid sample. Participants ranged in age from 21 to 48 years ($M = 29.5$), and the majority were female (22 female, 5 male). All participants had prior experience using LLM chatbots. Regarding mental health tool usage, 38% had never used any form of mental health support, while others had engaged with counseling (19%), counseling combined with chatbots (23%), chatbots only (8%), or online journaling systems

(12%). Our sample was predominantly female, which may limit the transferability of findings to other populations.

3.3 Chatbot System

We implemented a conversational AI system using GPT-4o as the base model. The system was designed with custom prompt engineering grounded in Social Support Theory [5], incorporating three types of support: emotional support (empathetic acknowledgment), appraisal support (affirmation of user strengths), and informational support (practical guidance). The system was explicitly framed as an “emotional support companion” rather than a clinical intervention. Response length was constrained to 1–3 sentences to encourage sustained conversational exchange. The prompt design was reviewed by five licensed mental health professionals to ensure it operated within safe boundaries and did not risk harm.

3.4 Procedure

Participants interacted with the chatbot daily for 10 consecutive days, with a minimum of 15 minutes per session. They could initiate conversations at any time and freely express emotions, report stress, or seek support. Every two days, participants completed a brief questionnaire assessing their interaction experience. After the 10-day period, we selected 8 participants for 60–90 minute semi-structured in-depth interviews. Selection followed a theoretical sampling strategy, targeting participants with notably different experiences: those who felt highly supported, those who reported minimal empathy, and those with atypical support preferences.

3.5 Analysis

Interview transcripts were analyzed using grounded theory [9], following three coding stages: open coding to identify initial concepts, axial coding to establish relationships between concepts, and selective coding to develop thematic patterns. The finding reported in this paper, the relationship between social support gaps and AI evaluation, emerged inductively during the axial coding phase as a recurrent theme across participants. It was not a pre-specified research question but an unexpected pattern that warranted further exploration.

4 Findings

While the broader study included diary data, the findings reported in this paper are based primarily on grounded theory analysis of the post-study interviews, given space constraints and the exploratory focus of this poster.

4.1 Overview

Through grounded theory analysis of eight in-depth interviews, we observed a consistent pattern: participants’ evaluations of the AI emotional support chatbot varied systematically based on perceived gaps in their existing social support. Participants who described barriers to accessing human support tended to evaluate the AI more positively, while participants with strong, accessible support systems were more critical. Below, we illustrate these patterns using representative participant quotes.

4.2 Participants Experiencing Support Gaps: AI as “Filling a Void”

Five participants described barriers to accessing human support, including fear of burdening others, unsatisfying relationships, negative counseling experiences, and difficulty discussing certain topics. For these participants, the AI was often described as a valuable, judgment-free space.

Fear of burdening others. P4 explained: “I worry that my emotions will affect others.” She described the AI as “like a tree hollow. There are things I cannot tell even close friends, but with the AI, I do not have to worry about being judged or burdening anyone.” She further noted that this sense of safety allowed her to be more honest: “I can say things I would never say out loud to anyone else. It feels like a private space that belongs only to me.”

Unsatisfying relationships. P7’s closest friend often asked him for money, making emotional reliance difficult. He rated his friends’ support as “6 to 7 out of 10” but rated the AI 8 out of 10: “Sometimes it is even better than real people. It is available anytime.”

Negative past counseling. P1 reported a mismatch with a previous counselor that left her anxious. She found the AI provided unconditional positive regard: “Because it is a machine, it has no hidden agenda. The sincerity comes from its lack of humanity.”

Emotional cost of human support. P2 described human interaction as requiring “emotional cost.” With the AI: “Talking to the chatbot does not feel as stressful. With humans, I have to spend extra emotional energy.”

4.3 Participants with Adequate Support: AI as “Falling Short”

Three participants with strong support systems or low support needs evaluated the AI against a higher standard and found it lacking.

Strong professional support. P8 (counseling masters student) described her support network as strong, with a supportive partner and friends from counseling backgrounds. She stated: “My support system is strong enough. . . No matter what tool it is, it cannot compare to a real person.” The AI’s responses felt formulaic: “It keeps saying ‘I am sorry to hear that,’ which feels like a polite script.”

Preference for human depth. P5 had positive counseling experiences and described his counselor as a “Johari window.” He was clear about AI’s limits: “Being truly understood still requires a real person. I want human recognition, not recognition from a machine.” He elaborated that his counselor could “read between the lines” and ask follow-up questions that revealed deeper insights, something he felt the AI could not replicate: “The AI responds to what I say, but my counselor responds to what I mean.”

Low support needs. P3 and P6 both described low daily needs for emotional support. P3 noted: “I prefer being alone and recharging by myself.” P6 stated: “Some people might need it, but I do not.” Both gave moderate ratings but emphasized the AI was not personally useful.

Table 1: Participant characteristics and AI evaluation patterns

ID	Gender	Support Gap Description	AI Rating	Key Quote
P1	F	Negative counseling experience	Positive	"The sincerity comes from its lack of humanity."
P2	F	Emotional cost of human support	Positive	"Talking to the chatbot does not feel as stressful."
P3	F	Low support needs	Moderate	"I prefer being alone and recharging by myself."
P4	F	Fear of burdening others	Positive	"Like a tree hollow... a private space that belongs only to me."
P5	M	Strong counseling relationship	Critical	"The AI responds to what I say, but my counselor responds to what I mean."
P6	F	Low support needs	Moderate	"Some people might need it, but I do not."
P7	M	Unsatisfying friendships	Positive	"Sometimes it is even better than real people."
P8	F	Strong support network	Critical	"No matter what tool it is, it cannot compare to a real person."

4.4 Summary

4.5 Interpretation

The pattern suggests that participants did not evaluate the AI chatbot based solely on its technical performance or empathic quality. Rather, their evaluations appeared to be shaped by the reference standard against which they judged it. Participants experiencing support gaps compared the AI to having limited or no support, and often found it helpful. On the other hand, participants who already have adequate support compared the AI to the depth of human connection, and often found it insufficient. Notably, this pattern held regardless of participants' actual emotional needs or the frequency of their chatbot use; what mattered was not how much support they needed, but how accessible they perceived their existing support to be. This suggests that the same AI system can be experienced as "surprisingly helpful" or "disappointingly shallow" depending on who is using it. This observation aligns with our proposed "support gap" framing and raises questions about how user context should inform the design and evaluation of AI emotional support systems.

5 Discussion

5.1 Revisiting the Support Gap Framing

Our findings lend initial support to the "support gap" framing proposed in this paper. Participants did not evaluate the AI chatbot in isolation; rather, their evaluations appeared to be shaped by the reference standard against which they judged it. For participants with gaps in their existing social support, such as fear of burdening others, unsatisfying relationships, or difficulty discussing certain topics, the AI was compared to having limited or no support, and was often perceived as valuable. For participants with strong, accessible support systems, the AI was compared to the depth and authenticity of human empathy, and was more likely to be found lacking. Importantly, we do not interpret support gaps as user deficiencies. Rather, we suggest that users' prior experiences shape the evaluative lens through which they perceive AI, regardless of the system's objective capability and quality.

This pattern suggests that aggregate evaluations of AI emotional support systems may obscure important variations in user experience. A system that appears "moderately effective" on average may in fact be highly valuable for some users and largely unhelpful for others, depending on their social support contexts.

5.2 Questions for Design

Rather than offering definitive design recommendations, we raise several questions that emerged from our findings and warrant further exploration:

Should AI emotional support systems account for users' social contexts? If users with support gaps perceive AI differently from those with adequate support, should systems be designed to recognize or adapt to these differences? What would such an adaptation look like in practice? The following questions are not conclusions drawn from our data, but provocations that our findings invite for future inquiry. What are the ethical implications of positioning AI as a supplement for underserved users? Our observations indicate that participants who described gaps in their support access tended to evaluate the AI more positively, though the reasons for this pattern remain to be explored. This raises concerns worth examining: Is AI filling a genuine need, or does it risk becoming a "second-best" option that reinforces existing inequities in emotional care access?

In terms of the evaluation of support systems, we can ask: If user evaluations depend heavily on their pre-existing support contexts, should evaluation frameworks incorporate measures of users' pre-existing social support backgrounds? Should we report outcomes separately for different user groups?

5.3 Limitations and Future Work

This initial investigation has several limitations: (1) Exploratory findings emerged from eight interviews, and the small sample limits generalizability. (2) Social support contexts were characterized through self-report rather than validated instruments. (3) All participants were recruited from Taiwan and were predominantly female, which may limit transferability. (4) The 10-day study period may not capture longer-term perception changes.

Future research could: (1) validate this pattern quantitatively with larger, diverse samples using standardized social support measures; (2) investigate mechanisms underlying the support gap effect; and (3) explore whether AI systems that adapt to users' social contexts are perceived differently across cultural settings.

6 Conclusion

This paper presents an exploratory finding from a 10-day diary study with an LLM-based emotional support chatbot. Through in-depth interviews with eight participants, we observed a consistent pattern: users' evaluations of AI emotional support appeared to vary based on perceived gaps in their existing social support systems.

Participants with support gaps tended to view the AI as a valuable, judgment-free resource, while those with strong support networks were more critical.

We propose a “support gap” framing as one lens for understanding this variation: users’ evaluations of such AI systems are not based solely on system quality, but relative to their existing support experiences. This framing challenges the implicit assumption that AI emotional support can be assessed independently of user context.

We offer this observation as an empirical starting point, not a definitive conclusion. We invite the CHI community to consider how users’ social support contexts might inform the design, evaluation, and ethical deployment of AI emotional support systems.

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